MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6216

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, power output filter reactor, beam power

Outline	---	6-3 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	a	g1	k g3	h	h	a	g2	k g3	nc

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Rg1	Ik	Pp	Pg2	TE	Alt
Unit:	V	Vdc	Vdc	Vdc	v	Meg	mAdc	W	W	°C	ft
Maximum:	6.9	300	0	200	200	Note 1	110	10	2.2	210	Note 2
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	-3	100	---	---	---	---	---	---	---

GENERAL:

Qualification - Required
Reliable tubes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Heater current		0.4	II	If	1.10	1.30	A
1266	Total grid current	Rg1 = 0.5 Meg (see note 3)	0.4	II	Ic1	---	-2.5	µAdc
1256	Electrode current (1) (anode)		0.4	II	Ib	55	95	mAdc
1256	Electrode current (2) (anode)	Eb = 45 Vdc	0.4	II	Ib	50	---	mAdc
1306	Transconductance (1)		0.4	II	Sm	10,000	15,000	µmhos
1341	Power output	Eb = 200 Vdc; Ec1 = -6 Vdc; Esig = 4.2 Vac; Rp = 4,500 ohms	0.4	II	Po	2.5	---	W
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1336	Heater-cathode leakage	Ehk = ±200 Vdc	2.5	II	Ihk	---	50	µAdc
1256	Electrode current (screen)		2.5	II	Ic2	---	6.0	mAdc
1211	Insulation of electrodes		2.5	I	R	50	---	Meg
1256	Electrode current (3) (anode)	Eb = 300 Vdc; Ec2 = 200 Vdc; Ec1 = -21.5 Vdc	---	---	Ib	21	45	mAdc
1306	Transconductance (2)	Ef = 5.7 V	---	---	ΔSm Ef	---	10	%
1331	Direct-interelectrode capacitance	No shield	6.5	Code E	Cg1a Cin Cout	--- 11 5.5	0.44 15.5 7.9	pF pF pF
1246	Audio frequency noise	Rp = 2,000 ohms; Rg1 = 0.1 Meg; Rg2 = 1,600 ohms; Cg2 = 2.0 µF; Ecat = 150 mVac (see note 4)	---	---	---	---	---	---
1031	Low-frequency vibration	Ec1 = -20 Vdc; Rp = 2,000 ohms	6.5	I	Ep	---	300	mVac
1266	Grid emission	Ef = 6.9 V; Ec1 = -35 Vdc; Rg1 = 0.5 Meg (see note 5)	---	---	Ic	---	-4.0	µAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1041	Shock	450 G; Ehk = 200 Vdc (see note 6)	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G fixed frequency; F = 25 (min), 60 (max)	6.5	See note 7	---	---	---	---
---	Shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	500	mVac
1256	Electrode current (1) (anode)		---	---	Ib	50	---	mAdc
1336	Heater-cathode leakage		---	---	Ihk	---	100	µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: SUB SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.5 Vac; Ehk = 200 Vdc; Ec1 = Ec2 = Eb = 0	---	---	---	---	---	---	---	---
---	Heater-cycling life-test end point:									
1336	Heater-cathode leakage		---	---	---	---	Ihk	---	100	µAdc
1516	Stability life	Ehk = 200 Vdc; Rg = 0.5 Meg; TA = room	---	---	---	---	---	---	---	---
---	Stability life-test end point:									
1306	Transconductance (1) of individual tubes		---	---	---	---	ΔSmt	---	10	%
1521	Survival-rate life	Stability life-test conditions; TA = room	---	---	---	---	---	---	---	---
---	Survival-rate life-test end points (100 hours):									
1201	Short and discontinuity detection (inoperatives)		---	---	---	---	---	---	---	---
1306	Transconductance (1)		---	---	---	---	---	8,500	---	µmhos
1501	Intermittent life	Rk = 39; Rg1 = 0.5 Meg; Eb = Ec2 = 120 Vdc; Ec1 = 0; Ehk = 200 Vdc; TE = 210°C (min) (see note 8)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	---	-2.5	µAdc
1256	Electrode current (1) (anode)		---	---	1	3	Ib	50	---	mAdc
1306	Transconductance (1)		---	---	1	3	ΔSmt	---	15	%
1306	Transconductance (2)		---	---	1	3	ΔSmEf	---	15	%
1341	Power output		---	---	1	3	Po	2	---	W
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	75	µAdc
1211	Insulation of electrodes		---	---	1	3	R	40	---	Meg
---	Total defectives		---	---	3	6	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Intermittent life-test end points (1,000 hours):									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	---	-2.5	µAdc
1256	Electrode current (1) (anode)		---	---	1	3	Ib	50	---	mAdc
1306	Transconductance (1)		---	---	1	3	ΔSm_t	---	20	%
1306	Transconductance (2)		---	---	1	3	ΔSm_{Ef}	---	15	%
1341	Power output		---	---	1	3	Po	1.8	---	W
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	100	µAdc
1211	Insulation of electrodes		---	---	1	3	R	30	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---
1511	Cathode interface	Eb = 250 Vdc; Ec2 = 100 Vdc; Ec1 = -50 Vdc; Ehk = -200 Vdc; Rg1 = 0.5 Meg; TA = room (see notes 9 and 10)	---	---	---	---	---	---	---	---
---	Cathode interface life-test end points (500 hours)	Ef = 5.7 ±0.05 Vdc; Eb = Ec2 = 80 Vdc; Ec/Ib = 10 mAdc (see note 11)	---	---	1	3	Ri	---	20	Ohms
---	Cathode interface life-test end points (1,000 hours)	Ef = 5.7 ± 0.05 Vdc; Eb = Ec2 = 80 Vdc; Ec/Ib = 10 mAdc (see note 11)	---	---	1	3	Ri	---	25	Ohms

NOTES:

1. With cathode resistor bias, Rg1 may be 0.5 Meg (max); with fixed bias, Rg1 may be 0.1 Meg (max).
2. See reduced pressure (altitude) rating, and altitude, maximum peak voltage in the basic document.
3. This test to be performed at the conclusion of the holding period.
4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef Vdc	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Rg1 Meg	Rk Ohms
6.9	120	0	120	0.5	39

6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of a lot failure, the lot shall be rejected and the succeeding lots subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at minimum specified temperature at any position in the life-test rack.

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NOTES: -Continued

9. Cathode interface life test.

(a) Cathode interface life test shall be conducted for 1,000 hours.

(b) Cathode interface life-test acceptance shall be on the basis of the specified 500 and 1,000 hour life-test end-point limits.

A. Reduced hours life test.

(a) Eligibility for reduced hours life test is established when no first sample failure due to the regular 1,000-hour life test has occurred in the preceding 3-consecutive lots.

(b) Reduced hours life test shall be conducted for 500 hours and acceptance shall be based on the 500-hour life-test end-point limits. Two 500-hour life-test lot failures occurring in the last 3-consecutive lots shall result in loss of eligibility for reduced hours life testing.

(c) The life-test sample from the first lot accepted each month shall continue on life test for an additional 500 hours (1,000 hours total life-test time). Failure of this sample to meet the 1,000-hour life-test end-point limits shall result in loss of eligibility for reduced hours testing.

B. The life-test sample shall be read at the following times:

0 hours
500 + 48 hours
-24
1,000 + 48 hours (when in force)
-24

Additional reading periods may be used at the discretion of the tube manufacturer.

10. The life-test sample shall consist of 20 tubes and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample due to the failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed three. A life-test defect is defined as a failure to meet the life-test end-point limits as specified in the tube specification sheet.

11. Preheat approximately 5 minutes prior to testing using Ef = 5.7 V; other electrodes disconnected.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Navy - AS, OS, MC, CG
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Preparing activity:
Navy - EC

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